① A quiz consists of 3 multiple-choice questions each with five choices (A, B, C, D, and E), then 4 true or false questions. In how many ways can a student answer the quiz?

$$\underset{Q1}{5} \cdot \underset{Q2}{5} \cdot \underset{Q3}{5} \cdot \underset{Q4}{2} \cdot \underset{Q5}{2} \cdot \underset{Q6}{2} \cdot \underset{Q7}{2}$$

$$= \boxed{2{,}000}$$

② In the school cafeteria, students choose their lunch from 5 sandwiches, 3 salads, 2 desserts, and 4 drinks. How many different lunches are possible for a student who chooses exactly one sandwich, one salad, one dessert, and one drink?

$$\underline{5} \cdot \underline{3} \cdot \underline{2} \cdot \underline{4} = \boxed{120}$$

(3) At a certain university, student ID numbe consist of two letters (A through Z) followed four digits (0 through 9).

(A) If repetition of letters and digits is allowe how many possible student ID numbers are

$$\underline{26} \cdot \underline{26} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10}$$

$$= \boxed{6,760,000}$$

(B) If neither repetition of letters nor of digits allowed, how many possible student ID numbe are there?

$$\underline{26} \cdot \underline{25} \cdot \underline{10} \cdot \underline{9} \cdot \underline{8} \cdot \underline{7}$$

$$= \boxed{3,276,000}$$

(C) If the letters must be two different vowels (A, E, I, O, U) and repetition of digits is allowed, how many possible student ID numbers are there?

$$\underline{5} \cdot \underline{4} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} = \boxed{200,000}$$

(D) How many student ID numbers, allowing repetition of letters and digits, start with a C, contain exactly one 3, and end with an even digit?

$$\underset{C}{\underline{1}} \cdot \underline{26} \cdot \underset{3}{\underline{1}} \cdot \underline{9} \cdot \underline{9} \cdot \underset{\text{Even}}{\underline{5}} + \underline{1} \cdot \underline{26} \cdot \underline{9} \cdot \underset{3}{\underline{1}} \cdot \underline{9} \cdot \underset{\text{Even}}{\underline{5}}$$

$$+ \underset{E}{\underline{1}} \cdot \underline{26} \cdot \underline{9} \cdot \underline{9} \cdot \underset{3}{\underline{1}} \cdot \underset{\text{Even}}{\underline{5}} = \boxed{31,590}$$

can draw out three different cards from a standard deck of 52 cards and line them up in the order in which you drew them. How many possible lineups are there so that:

(A) The first card is a club and the second card is a red King?

$$\underset{\text{Club}}{13} \cdot \underset{\text{Red King}}{2} \cdot 50 = \boxed{1,300}$$

(B) All three cards are Queens?

$$4 \cdot 3 \cdot 2 = \boxed{24}$$

OR

$$P(4,3) = \boxed{24}$$

(C) None of the three cards is a club?

$$39 \cdot 38 \cdot 37 = \boxed{54,834}$$

OR

$$P(39,3) = \boxed{54,834}$$

(D) The Ace of Diamonds is in the lineup?

$$\underset{A\diamondsuit}{1} \cdot 51 \cdot 50 + 51 \cdot \underset{A\diamondsuit}{1} \cdot 50 + 51 \cdot 50 \cdot \underset{A\diamondsuit}{1}$$

$$= \boxed{7,650}$$

(E) At least one Jack is in your lineup?

$$\underbrace{P(52,3)}_{\text{All lineups}} - \underbrace{P(48,3)}_{\text{Lineups with no Jacks}} = 132,600 - 103,776 = \boxed{28,824}$$

(5) You roll a green die and a purple die. Both dice are standard 6-sided dice. You roll each once and record which die shows which value. How many possibilities are there for which the green die shows a number greater than or equal to 5 or the purple die shows a number less than or equal to 4?

$$\underbrace{\frac{2}{G}\cdot\frac{6}{R}}_{\text{Green Die Shows} \geq 5} + \underbrace{\frac{6}{G}\cdot\frac{4}{R}}_{\text{Red Die Shows} \leq 4} - \underbrace{\frac{2}{G}\cdot\frac{4}{R}}_{\text{Green} \geq 5 \text{ and Red} \leq 4}$$

$$= \boxed{28}$$

1 2 3 5 Even

⑥ A pianist plans to play 6 pieces at a recital from her repertoire of 25 pieces, and is carefully considering which song to play first, second, etc to create a good flow. How many different recital programs are possible?

$$P(25,6) = \boxed{127{,}512{,}000}$$

⑦ Tina signed up to bring 3 bags of chips to her work's party. She goes to the store and finds a total of 21 different ~~types~~ types of chips. If she wants to choose 3 different types of chips to the party, how many ways can she do this

$$C(21,3) = \boxed{1{,}330}$$

(8) You are dealt three different cards from a standard deck of 52 cards. How many possible hands (order does not matter) are there so that:

(A) Exactly two cards are hearts or exactly one card is a black Ace?

$$\underbrace{\underbrace{C(13,2)}_{\heartsuit}\ \underbrace{C(39,1)}_{\text{Non-}\heartsuit}}_{\text{Two Cards Hearts}} + \underbrace{\underbrace{C(2,1)}_{\text{Black Ace}}\ \underbrace{C(50,2)}_{\text{Non-Black Ace}}}_{\text{1 Black Ace}} - \underbrace{C(13,2)\cdot C(2,1)}_{\substack{\text{Two }\heartsuit \\ \text{and 1} \\ \text{Black Ace}}}$$

$$= \boxed{5{,}336}$$

(B) One card is a Jack, one is a Queen, and one is a King?

$$\underbrace{C(4,1)}_{\text{Jack}} \cdot \underbrace{C(4,1)}_{\text{Queen}} \cdot \underbrace{C(4,1)}_{\text{King}} = \boxed{64}$$

(C) All three cards are Aces?

$$C(4,3) = \boxed{4}$$

(D) At least one of the cards is a spade?

$$C(52,3) - C(39,3) = \boxed{12{,}961}$$

~~No spades~~

$C(52,3)$: All possible 3-card hands

$C(39,3)$: 3-card hands without spades

(9) In a survey of videogamers, it was found that 23 people play Minecraft and 44 people play Fortnight. If 15 people play both video games, how many people play Minecraft or Fortnight?

$$23 + 44 - 15 = \boxed{52}$$

(10) You are buying cookies at a bake sale. The five types of available cookies are: chocolate chip, sugar, M&M, oatmeal, and mint chocolate. You are going to buy three cookies. The order you choose the cookies does not matter, and you can choose any flavor multiple times to get your three cookies. How many possibilities are there for your three cookies?

$$C(5+3-1, 3) = C(7,3) = \boxed{35}$$

(11) You have a collection that consists of 6 green marbles, 6 black marbles, 2 blue marbles, 3 purple marbles, and 1 clear marble. Marbles of the same color are indistinguishable. How many ways can you line up (order matters) the marbles to show off your collection?

$$n = 6+6+2+3+1 = 18$$

$$p_1 = 6$$
$$p_2 = 6$$
$$p_3 = 2$$
$$p_4 = 3$$
$$p_5 = 1$$

$$\frac{18!}{6!6!2!3!1!} = \boxed{1,029,188,160}$$

(12) At a certain university, there are exactly 6 student clubs: math, physics, chess, astronomy, philosophy, and a book club. The clubs have combined their resources to throw an end-of-semester party. At least how many students (from the clubs) must attend the party to ensure at least 2 of them belong to the same club?

$\boxed{7}$

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